

NLP — 10-Page Master Quick Revision

High-Yield Formula Sheet, Trellis Trace Matrices & Top 10 BIT Mesra PYQ Solutions

⚡ 10-Page Master Quick Revision — Natural Language Processing (CS24311)

Page 1: Linguistic Levels, Byte-Pair Encoding & Levenshtein Edit Distance	Page 2: N-gram Language Models, Perplexity & Kneser-Ney Smoothing Formulations
Page 3: POS Tagging: Penn Treebank Tagset & Hidden Markov Models (HMM)	Page 4: The Viterbi Dynamic Programming Algorithm & Trellis Tracing Math
Page 5: Syntactic Parsing: CFG, Chomsky Normal Form (CNF) & CYK Dynamic Algorithm	Page 6: Dependency Syntax: Shift-Reduce, Arc-Standard & Arc-Eager Parsing
Page 7: Vector Space Models: TF-IDF, Positive PMI & WordNet Distance Metrics	Page 8: Word2Vec: Skip-Gram with Negative Sampling (SGNS) Loss & GloVe Matrix Model
Page 9: The Transformer Architecture: Scaled Dot-Product & Multi-Head Attention	10: BERT vs. GPT Architectures & Machine Translation Evaluation (BLEU / ROUGE)

⚡ Master Formula, Algorithm & Loss Function Cheat Sheet

Topic / Model	Core Mathematical Formulation / Rule	Key Exam Takeaway
Perplexity (PPL)	$\text{PPL}(W) = 2^{-\frac{1}{n} \sum_{i=1}^n \log_2 P(w_i w_{i-1})}$	Lower is better; represents average branching factor of language model.
Viterbi Recurrence	$v_t(j) = \max_i [v_{t-1}(i) \cdot a_{ij}] \cdot b_j(o_t)$	Finds globally optimal hidden state sequence in $O(N^2T)$ time.
CYK Parsing Recurrence	$P[i, j, A] = \bigvee_{k=i}^{j-1} \bigvee_{A \rightarrow BC} (P[i, k, B] \wedge P[k + 1, j, C])$	Requires grammar strictly in Chomsky Normal Form (CNF); runs in $O(n^3 P)$.
Skip-Gram SGNS Loss	$\mathcal{L} = \log \sigma(\mathbf{v}'_c \cdot \mathbf{v}_w) + \sum_{i=1}^k \mathbb{E}[\log \sigma(-\mathbf{v}'_{n_i} \cdot \mathbf{v}_w)]$	Maximizes true context dot product while pushing away k negative noise samples.
Scaled Dot-Product Attention	$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$	Scaling by $\frac{1}{\sqrt{d_k}}$ prevents vanishing softmax gradients.
BLEU Brevity Penalty	$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r \end{cases}$	Penalizes artificially short candidate translations that inflate precision.

🔥 Top 10 High-Yield BIT Mesra Exam Questions & Solutions

Q1. Given Query Q , Key K , Value V , trace the dimensions of the Transformer Multi-Head Attention operation with $d_{\text{model}} = 512, h = 8$. (8 Marks)

1. Dimension per head: $d_k = d_v = \frac{d_{\text{model}}}{h} = \frac{512}{8} = 64$.
2. Projections: $QW_i^Q \in \mathbb{R}^{n \times 64}, KW_i^K \in \mathbb{R}^{m \times 64}, VW_i^V \in \mathbb{R}^{m \times 64}$.
3. Attention Matrix: $\frac{Q_i K_i^T}{\sqrt{64}} \in \mathbb{R}^{n \times m}$. Softmax normalizes each row into attention weights.
4. Head Output: $\text{head}_i = \text{softmax}(\dots) V_i \in \mathbb{R}^{n \times 64}$.
5. Concatenation: $\text{Concat}(\text{head}_1 \dots \text{head}_8) \in \mathbb{R}^{n \times 512}$. Multiplying by $W^O \in \mathbb{R}^{512 \times 512}$ produces final output $\mathbb{R}^{n \times 512}$.